Cloud Computing & DevOps – AWS Handbook

SESSION 7

Docker Fundamentals

Introduction to Containerization

Duration	2 Hours
Prerequisites	Linux Fundamentals, EC2 Knowledge
Learning Outcome	Understand containers, images, Docker architecture, and deploy applications using Docker containers.

Why Are We Learning This?

Modern applications are increasingly deployed using containers instead of traditional virtual machines. Containers provide portability, consistency, faster deployment, and efficient resource utilization. Docker is the most widely adopted containerization platform and forms the foundation of modern DevOps practices.

Architecture Diagram

```
Application Code
↓
Docker Image
↓
Docker Container
↓
Host Operating System
↓
Infrastructure
```

Key Concepts

Container	Lightweight isolated execution environment.
Docker Engine	Core service that manages containers.
Docker Image	Blueprint used to create containers.
Docker Container	Running instance of an image.
Dockerfile	File containing image build instructions.
Container Registry	Repository used to store images.

Practical Activities

• Install and verify Docker

• Run first Docker container

• Explore Docker images

• Create a Dockerfile

• Build a custom Docker image

• Deploy a web application inside a container

• Access the application through a browser

Expected Outputs

Docker installed successfully, containers running, custom image created, and application accessible through a web browser.

What You Should See

Paste Screenshot Here

Docker Version Verification

Paste Screenshot Here

Docker Images List

Paste Screenshot Here

Running Containers

Paste Screenshot Here

Docker Image Build Output

Paste Screenshot Here

Application Running in Browser

Journal Writing Template

Aim

To understand Docker containerization and deploy an application inside a Docker container.

Theory

Docker provides containerization technology that packages applications along with their dependencies.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Docker was installed successfully, images were created, and containers executed correctly.

Conclusion

Docker simplified application deployment by packaging applications into portable containers.

Viva Preparation

1. What is Docker?
2. What is Containerization?
3. Difference between Containers and Virtual Machines?
4. What is a Docker Image?
5. What is a Docker Container?
6. What is a Dockerfile?
7. Why is Docker important in DevOps?

Troubleshooting Box

Problem	Possible Cause
Container not starting	Incorrect image or configuration
Port not accessible	Port mapping issue
Image build failed	Dockerfile error
Application unavailable	Application service not running

Industry Relevance

Docker is the standard container platform used across cloud providers and enterprise environments. It enables consistent deployment from development to production and is a prerequisite for Kubernetes and modern DevOps workflows.

Session Summary

Students learned Docker fundamentals, containerization concepts, image creation, container deployment, and application hosting using Docker.